Answer Key - Mathematics - Undergraduate Level Assessment 17 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** B
   Options: A) 1; B) 2; C) infinitely many; D) 0
   Note: There are 2 homomorphisms from Z to Z_2, corresponding to the two possible mappings of the generator 1 in Z to either 0 or 1 in Z_2, as any homomorphism must respect the operation and the structure of the groups involved.

2. **Answer:** C
   Options: A) subgroup; B) finite abelian group; C) infinite, non abelian group; D) ininite, abelian
   Note: The set G of all n x n non-singular matrices with rational entries forms an infinite group under multiplication because it contains infinitely many matrices, and it is non-abelian due to the fact that matrix multiplication is not commutative in general.

3. **Answer:** C
   Options: A) 2; B) 3; C) 4; D) 12
   Note: The order of the factor group (Z_4 x Z_12)/(<2> x <2>) is 4 because the subgroup <2> x <2> has an order of 4, and the order of the group Z_4 x Z_12 is 48, leading to a factor group with order 48/12 = 4.

4. **Answer:** D
   Options: A) pair of points; B) circle; C) half-line; D) line
   Note: The equation $z^2 = |z|^2 impliesthattherealpartandimaginarypartofzmustsatisfyalinearrelati$
   $plane$.

4. **Answer:** D
   Options: A) True, True; B) False, False; C) True, False; D) False, True
   Note: The symmetric group S_3 is not cyclic because it cannot be generated by a single element, while it is true that every group is isomorphic to a subgroup of some permutation group, as established by Cayley's theorem.

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: The reason this statement is true is that a homomorphic image of a cyclic group is generated by the image of a single generator of the original group, thereby preserving the cyclic structure.

7. **Answer:** True
   Options: A) True; B) False

Note: A function from a finite set onto itself is one-to-one because if it were not, at least two elements would map to the same element, violating the requirement that every element in the codomain must be hit by the function, thereby preventing the function from being onto.

8. **Answer:** False
   Options: A) True; B) False
   Note: The product of the polynomials f(x) = 4 x - 5 and g(x) = 2 x^2 - 4 x + 2 in Z_8[x] is calculated as (4 x - 5)(2 x^2 - 4 x + 2) = 8 x^3 - 16 x^2 + 8 x - 10, which simplifies to 0 in Z_8[x] because all coefficients are reducible to zero modulo 8, making the statement false.

9. **Answer:** False
   Options: A) True; B) False
   Note: The maximum possible order of an element in the direct product Z_8 x Z_10 x Z_24 is determined by the least common multiple of the orders of the individual groups, which is 120, not 240.

10. **Answer:** False
    Options: A) True; B) False
    Note: The statement is false because a group G can satisfy the property $(ab)^2 = a^2\ b^2$ for all a, b in G without being of order two, as this condition is satisfied by any abelian group, not just those of order two.

Long Form Response

11. **Answer:** Let $Q = (x, y)$. From the given information, $y^2 = 4 - 4x^2$. Therefore, PQ^2
    &= (x + 1)^2 + y^2
    &= x^2 + 2 x + 1 + 4 - 4 x^2
    &= -3 x^2 + 2 x + 5
    &= -3 (x - $\frac{1}{3}$)^2 + $\frac{16}{3}$. This is maximized when $x = \frac{1}{3}$, and $d^2 = \frac{16}{3}$.
    Note: The answer correctly identifies the distance formula between points P and Q, substitutes the constraint of the ellipse to express d^2 as a quadratic function of x, and finds the maximum value using the vertex formula, yielding $d^2 = \frac{16}{3}$.

12. **Answer:** A linear transformation P on a finite-dimensional real vector space V such that $P^2 = P$ is known as a projection. One of the key implications of this property is that P is idempotent, meaning applying the transformation twice yields the same result as applying it once. While projections are not necessarily diagonalizable in all vector spaces, they can be shown to be diagonalizable if we consider the eigenvalues and their corresponding eigenspaces. Specifically, the only eigenvalues of P are 0 and 1, leading to a decomposition of V into the direct sum of the eigenspaces associated with these eigenvalues. Thus, P is indeed necessarily diagonalizable in the context of real vector spaces, confirming that P can be expressed in a diagonal form with respect to a suitable basis.

Note: correct because it identifies that a linear transformation P satisfying $P^2 = P$ is a projection, which is idempotent, and discusses the conditions under which such a transformation can be diagonalizable by examining its eigenvalues and eigenspaces in finite-dimensional real vector spaces.

End of Answer Key